Vibecoding and AI usage

CRI

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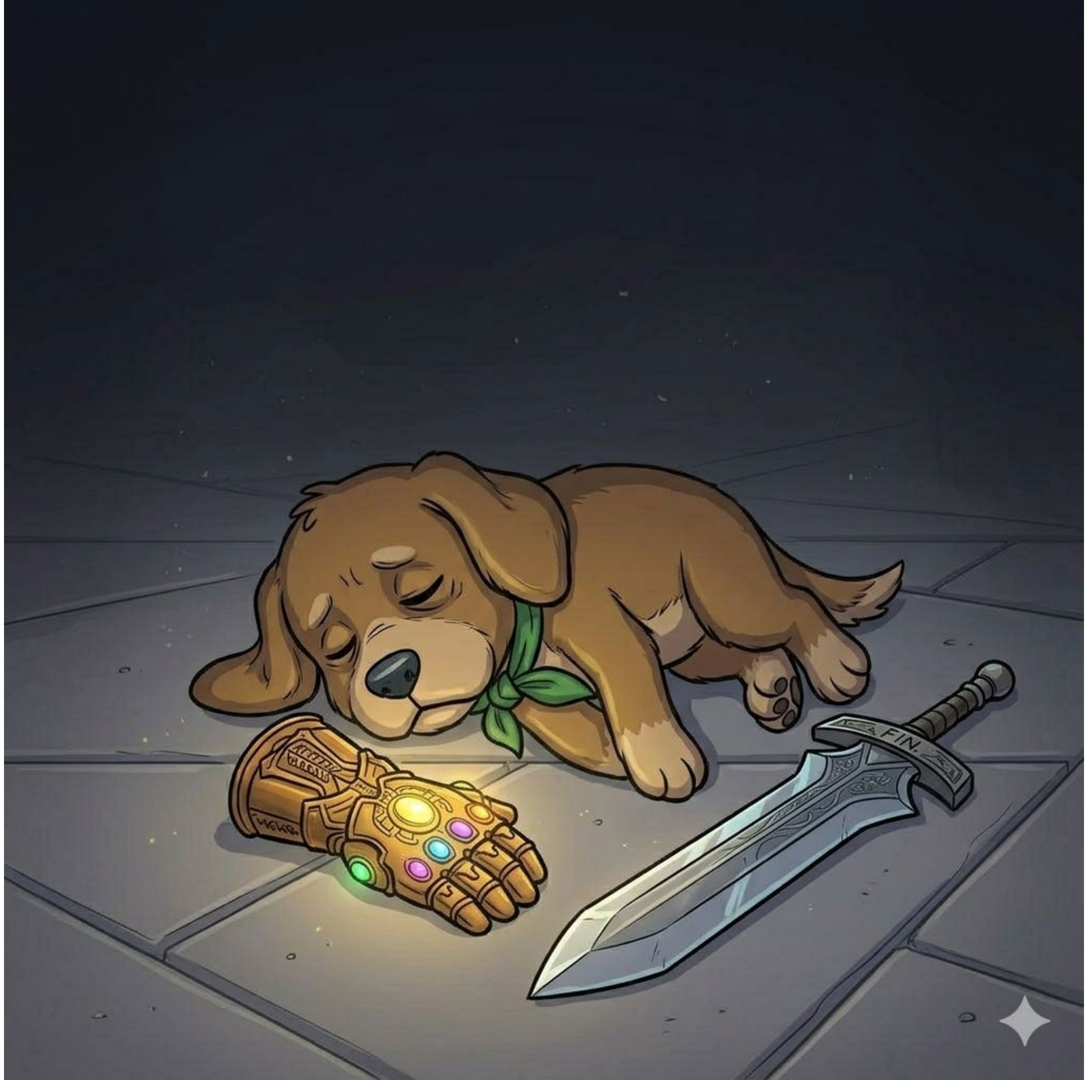
This week …

- Introduce vibecoding, or coding with help of an LLM/Chatbot.
- How does LLM use the code it suggests?
- Best Practices
- What NOT to do
- What can go wrong?
- But more importantly …

Dog needs a rest,
he will be back
next week



Vibecoding?

What is it?

Students: Let’s be honest, everyone here has vibecoded.

Students: Let’s be honest. You have vibecoded in this course, and pretended you know what you are doing without really understanding what you are doing.

Staff, let’s be honest: We know when you vibecode, and if we wanted, we could fail you for dishonest practice.

As an experienced programmer, it is really easy to identify the difference between code generated by a human and code generated by an AI.

This lecture is not about demonising vibecoding.

This lecture is about helping you use it ethically to make you a better programmer, a better creative, and, overall, a better practitioner.

Vibecoding, what is it?

- The term was coined by Andrej Karpathy, an AI researcher.
- Originally, a term used to describe telling an AI what to do and letting it write almost all the code with little to no human intervention.
- Later, it was adopted to describe the usage of AI and LLM to help developers write code.

Vibecoding in the wild

- Because AI is not going anywhere, we need to learn how to use vibecoding in our programming projects.
- More and more developers start to implement vibecoding in their practice.


VIBE CODING
EVERYTHING
LIFE & CODE
IDEAS
VIBES
CODE
DESIGN
PIXELS
FUNCTIONS...
COFFE


IOGAS
CODE
DESIGN
PIXELS
REFACTOR
API
SSCALING
BUGS
DEADLINES
REQUIREMENTS
DEPLOYMENT
SCALING
LEGACY CODE
COFFFEE
IDEAS
VIBES
TODAY'S BUG-LOG


LETS VIBECODE
RESPONSIBLY

But how does vibecoding work?

How does an LLM generate the code I copy paste everywhere

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How does LLM generate code?

- First of all, think of any LLM, ChatGPT, Claude, Gemini, Deep Seek, etc.
- They do not think a problem the same way humans do; they are not that smart (I hope).
- These algorithms ***do not understand what you want to create.***
- They are playing a very sophisticated game of ***"What word (or line of code) probably comes next?"***
- Let's see an example

How LLMs (kind of) work

How LLMs (kind of) work

LLM Space

Human Space

How LLMs (kind of) work

LLM Space

Human Space

I want to create a for loop

How LLMs (kind of) work

LLM Space

There is a 85% chance they want me to start by typing the keyword “for”

Human Space

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How LLMs (kind of) work

<u>LLM Space</u>

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<u>Human Space</u>

I want to create a for loop

for (….

How LLMs (kind of) work

<u>LLM Space</u>

There is a 85% chance they want me to start by typing the keyword “for”

Statistically, the most likely next part after for is the iterator.

<u>Human Space</u>

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Human Space

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for (….

for (let I = 0; I < limit; i++)

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for (let I = 0; I < limit; i++)

OMG you are so smart

How LLMs (kind of) work

<u>LLM Space</u>

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Convince human to give more money to overlord Sam Altman

<u>Human Space</u>

I want to create a for loop

for (….

for (let I = 0; I < limit; i++)

OMG you are so smart

Does anyone see a problem in here?

Consider that LLM was trained with billions of lines of code, including multiple interpretations of for loops.


The internet is full
fo garbage code


The internet is full of garbage code!

Problem leap year calculation

deepthi.kodakandla · May 29, 2024

deepthi.kodakandla
Joined: May 29, 2024
Messages: 1
Reaction score: 0

May 29, 2024 #1

Leap year logic. Is it correct? Basic rule of leap year rule is it comes once every 4 years .
everywhere online when checking the leap years between 1695 and 1705 its showing as 2 and the years are 1696 and 1704 which are having 8 years of gap between them. it's happening between 1795 and 1805 too .its showing 1796 and 1804 which are again having 8 years of gap. the logic is giving this kind of gap between around 1900 too . its is showin 1896 and 1904 as leap years . again gap of 8 years to get a leap year . time to change the logic which is incorrect

Reply

WhiteCube
Joined: Sep 20, 2022
Messages: 284
Reaction score: 41

May 29, 2024 #2

Leap, if divisible by 4, but there's an exception to that (Y%100) and an exception to the exception (Y%400).

Code:

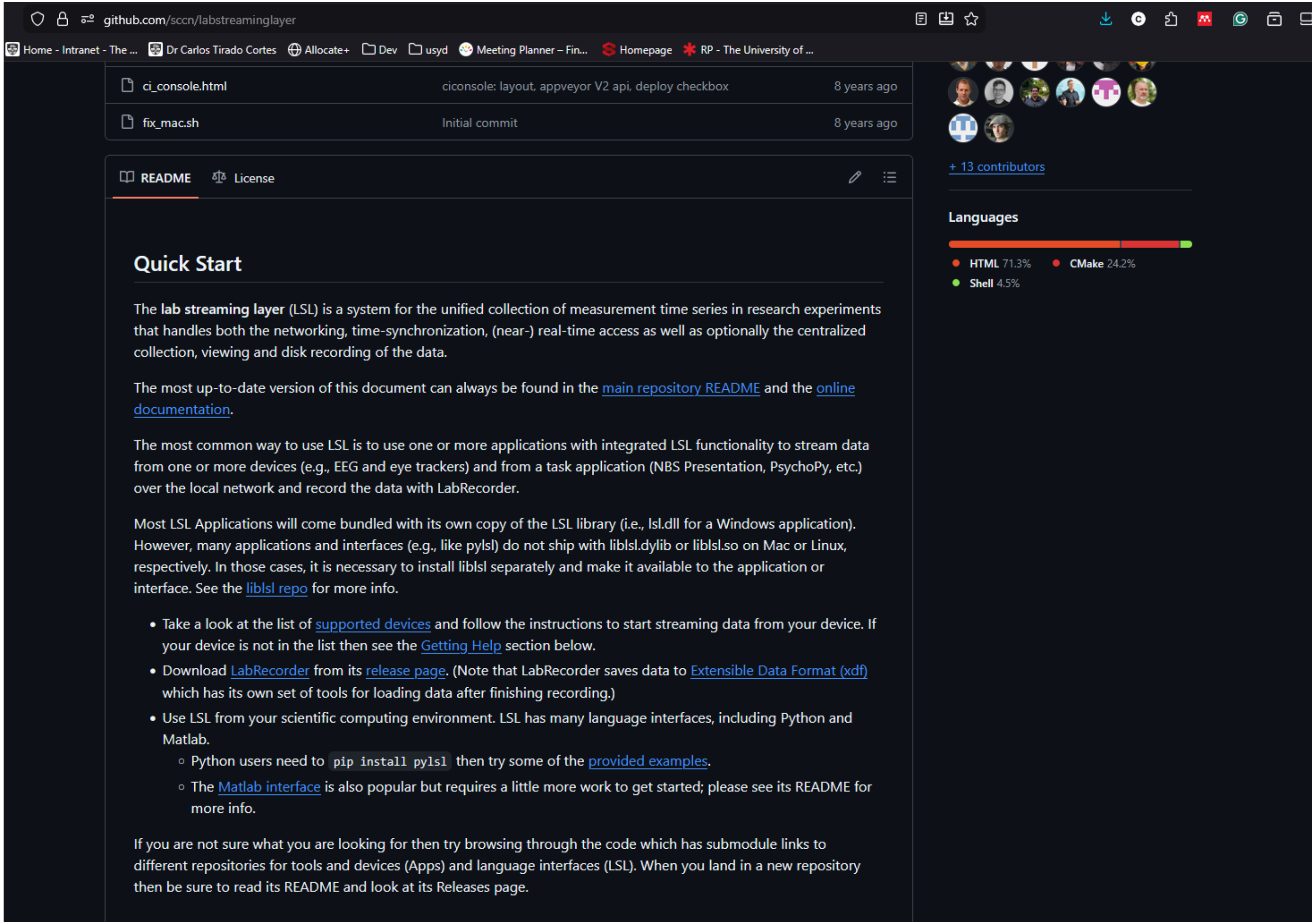
```
pseudocode

FUNCTION IS_LEAP(Y)
L=FALSE
IF (Y%4)==0
  L=TRUE
IF (Y%100)==0
  L=FALSE
IF (Y%400)==0
  L=TRUE
RETURN L
```

Reply

There is some good code out there, too!

Google: Lab Streaming Layer



github.com/sccn/labstreaminglayer

Home - Intranet - The ... Dr Carlos Tirado Cortes Allocate+ Dev usyd Meeting Planner – Fin... Homepage RP - The University of ...

ci_console.html	ciconsole: layout, appveyor V2 api, deploy checkbox	8 years ago
fix_mac.sh	Initial commit	8 years ago

README License

Quick Start

The **lab streaming layer** (LSL) is a system for the unified collection of measurement time series in research experiments that handles both the networking, time-synchronization, (near-) real-time access as well as optionally the centralized collection, viewing and disk recording of the data.

The most up-to-date version of this document can always be found in the main repository README and the online documentation.

The most common way to use LSL is to use one or more applications with integrated LSL functionality to stream data from one or more devices (e.g., EEG and eye trackers) and from a task application (NBS Presentation, PsychoPy, etc.) over the local network and record the data with LabRecorder.

Most LSL Applications will come bundled with its own copy of the LSL library (i.e., lsl.dll for a Windows application). However, many applications and interfaces (e.g., like pylsl) do not ship with liblsl.dylib or liblsl.so on Mac or Linux, respectively. In those cases, it is necessary to install liblsl separately and make it available to the application or interface. See the liblsl repo for more info.

- Take a look at the list of supported devices and follow the instructions to start streaming data from your device. If your device is not in the list then see the Getting Help section below.
- Download LabRecorder from its release page. (Note that LabRecorder saves data to Extensible Data Format (xdf) which has its own set of tools for loading data after finishing recording.)
- Use LSL from your scientific computing environment. LSL has many language interfaces, including Python and Matlab.
  - Python users need to `pip install pylsl` then try some of the provided examples.
  - The Matlab interface is also popular but requires a little more work to get started; please see its README for more info.

If you are not sure what you are looking for then try browsing through the code which has submodule links to different repositories for tools and devices (Apps) and language interfaces (LSL). When you land in a new repository then be sure to read its README and look at its Releases page.

+ 13 contributors

Languages

HTML 71.3% CMake 24.2% Shell 4.5%

Problem: An LLM can’t tell the good from the bad code. It will just assess the most common patterns and try to replicate them.

For simple code like that of Weeks 1 and 5, you can get away with vibecoding very easy.
For a niche topic like creative coding?
It will be a roll of the dice.

~~Douglas~~ Carlos Tirado 2026

What happens if you ask ChatGPT to help you with the final assessment?

- GPT will stitch together code from multiple places and try to make it work.
- No matter how good your prompt is, GPT will be constantly guessing what you want.


?
FRANKENSTEIN CODE -
- A MONSTER OF OUR OWN MAKING

Most common mistakes students do when submitting projects that used AI to help them code

AI used very complex code that is out of the scope of the unit. Because the student does not understand what is happening, they just assume the answer is right.

The student failed the assessment due to excessive complexity in the code.

In my years of coding, I’ve never seen code like this

Jared Berghold, 2025, during parity meeting

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Overcomplicated code that works but adds unnecessary complexity, lowering the quality of the output.

Code that works on their computer but does not work anywhere else.

Mix-and-match of different coding styles, making the project look like someone wrote the code for the student.

What about creative coding?

Issues with P5.js, Processing and Creative Coding Tools

- There are way fewer creative coding projects on the internet than full JavaScript or Python projects.
- There are even fewer creative and artistic examples that can help out.
- Some artist put their code behind a paywall, claiming that their code is part of the IP.

P5.js and Creative Coding

Common issues with AI-assisted p5.js coding

1 **Hallucinated or misused functions**

Invents p5.js functions that don't exist, or calls real ones incorrectly

2 **Library confusion**

Mixes in Processing or Three.js syntax — looks right, but it won't run

3 **Works — but not what you actually wanted**

You spend 3 hours tweaking it to match your vision...

...only to realise you could've just coded it from scratch in that time

Vibecoding: Best Practices

Best Practices

- Start Small and Build Up
- Context is key
- Always read and understand before using
- Get help with repetitive boilerplate code
- Test, and Test Incrementally
- Debugging is a Trap!

Start Small, Then Build Up

Ask for help in specific chunks

❌ Too Big

Build an entire sketch with animation, collision detection, and three different shape types

→ **Hallucinations** 🤖

✓ Just Right

Create a function that calculates **square root**

↓

Create a function that generates **random colors**

↓

Create a function that detects **mouse clicks**

Give Context to Your Prompt

Tell the LLM exactly what concepts you know

Your Prompt

Topics I've learned:

- Variables
- Loops
- Conditionals
- Functions

Libraries I use:

- p5.js v1.7

My goal:

"Create a simple 3-line function that..."

Be specific about what you know

What LLM Returns

LLM Might Use:

❌ Classes

❌ Arrow functions

❌ setTimeout()

❌ Async/await

❌ Canvas transforms

Why?

LLMs hallucinate confident features even when told to avoid them 🤖

Always Read and Understand Before Using

Ask the LLM to explain the code

❌ Don't	⚠️ Risky	✓ Better
Copy code directly into your project without reading it first **↓ You won't learn anything**	Ask LLM to debug it for you without full context **↓ LLM guesses wrong**	Ask LLM: **"Explain this code step by step"** Or: **"Can you make it simpler?"** **↓ You learn!**

Use it for Repetitive and Boilerplate Code

Where experienced developers find real value

Scenario: Create 5 similar objects with different values

You need to assign unique IDs incrementally to 5 character objects

❌ Manual (Tedious)

```
let character  = {
    id:  1 ,
    name:   "spiderman"
}

let character2  = {
    id:  2 ,
    name:   "daredevil"
}
```

Tedious! Error-prone!

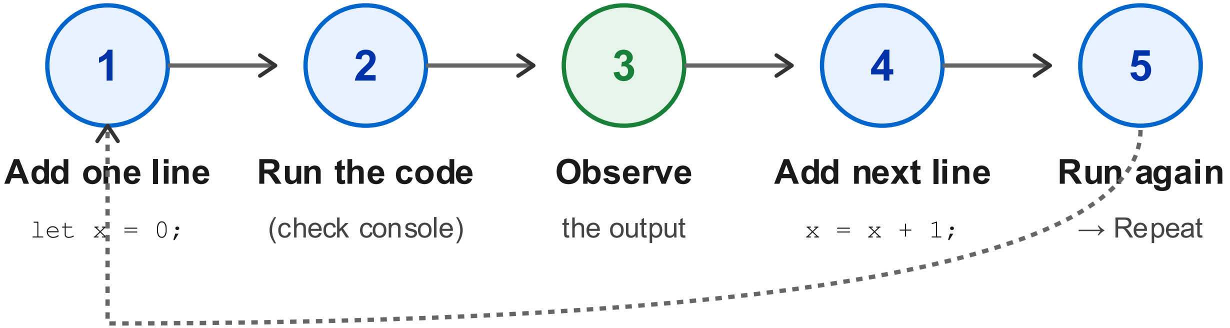
✓ Ask AI for Pattern

```
letheroNames = ["spiderman" ,"daredevil" ,
"hawkeye" ]
letheroes
for(leti= 0 ; i <= heroNames.length ; i ++) {
  heroes [i ] = {
  id: i + 1 ,
  name: heroNames[i ]
  }
}
```

Smart use of AI! ✓

Test and Test Incrementally

Add line by line, run after each addition



Why is this better?

✓ You understand what each line does

✓ Errors appear immediately, not later when debugging

Debugging is a Trap

Don't ask the LLM to fix broken code—ask it to explain the error

❌ The Trap

You ask:

"Why doesn't my code work?"

LLM searches web + guesses solutions

Problem:

LLM missing YOUR full context

✓ Be a Detective

Step 1: Check your

CONSOLE

for error message

Step 2: Ask LLM:

"Why would this error occur?"

Result:

YOU become the problem solver

What not to do

List of not very good practices

- Blindly trust the code
- Don’t ask an LLM to solve a huge problem in one go
- Don’t use Concepts/Functions that you haven’t learned yet
- DON’T LET IT MAKE YOUR CREATIVE DECISIONS
- Do not substitute reading the documentation for help from an LLM
- AI is never a substitute to learn code

Don't Blindly Trust the Code

Confidence does not mean correctness

LLM output (looks right)

```
function drawScene() {
  let layer = canvas
    .getActiveLayer();
  layer.renderAll(
    { autoRefresh: true });
  p5.applyFilter(
    'glow', 0.8);
}
// totally legit... right?
```

but...

Reality check

✗ .getActiveLayer()
Not a p5.js function

✗ .renderAll()
Does not exist

✗ p5.applyFilter()
Made-up syntax

All stated with full confidence

Always double-check before implementing anything an LLM gives you

If you cannot find it in the p5.js docs, it probably does not exist

Don't Ask AI to Solve the Entire Assessment

You will learn nothing — and the mismatch is obvious

What you know at this stage

✓ Variables and constants
✓ If / else statements
✓ For loops
✓ Functions
✓ Arrays

Honest, in-scope code

What AI may produce instead

✗ Inheritance and Polymorphism
✗ Design patterns
✗ Async / await
✗ Higher-order functions
✗ Recursive algorithms

...worthy of a CS graduate

Over-complex code is obvious — students are surprised when they fail

Final assessment: 1 mark deducted per out-of-scope line you cannot explain

Don't Use Concepts You Haven't Learned Yet

Even when you specify constraints, LLMs often ignore them

Student prompt

"Can you solve this using
only for-loops and if statements?"

LLM response

`for (let i = 0; i < arr.length; i++)`	✓ OK
`if (arr[i] > max) { max = arr[i]; }`	✓ OK
`arr.map(x => x * 2)`	✗ Not asked for
`arr.filter(Boolean)`	✗ Not asked for
`class DataProcessor { constructor() { ... } }`	✗ Out of scope

Tip: Write "ONLY USE for-loops and if statements" — then always verify the output

If you don't recognise a function, ask GPT to explain it before using it

Don't Let AI Make Your Creative Decisions

LLMs are trained on what exists — they cannot imagine what doesn't

Prompt

"Make something cool with circles"

AI produces

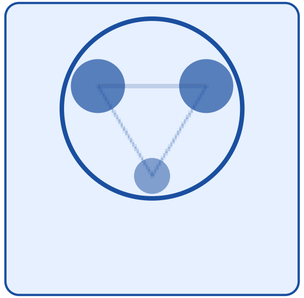
Generic. Forgettable.
No concept. No YOU.

This is a Creative Coding class.

Part of what you are here to learn
is making creative decisions
that you can turn into code.

AI is here to help with the technical parts.

Your vision



Intentional. Conceptual.
Your idea, executed.

Check the Inspirations section — then compare it with what AI generates.

Develop and express YOUR ideas. AI executes — you imagine.

Don't Substitute the p5.js Docs with an LLM

Always double-check that the code you receive actually exists

LLM says...

Looking for a blur filter?

```
p5.Canvas.setFilter(
'blur', 5)
```

Need raw pixel data?

```
canvas.getPixelArray()
```

Want to blend colours?

```
p5.colorMix(c1, c2, 0.5)
```

Sounds legit. Doesn't exist.

vs

p5.js docs say...

Looking for a blur filter?

```
filter(BLUR, 5)
```

Need raw pixel data?

```
loadPixels() / pixels[]
```

Want to blend colours?

```
lerpColor(c1, c2, 0.5)
```

Verified. Always works.
Always up to date.

p5.js is constantly updated — your LLM may not know the latest version

When in doubt: p5js.org/reference — the ground truth, always

AI is Never a Substitute to Learn Code

It will make your learning slower or incomplete

Shortcut path

Prompt AI → Copy paste

↓

Submit without understanding

↓

Code doesn't work. Panic.

Skill level stays flat.

The learning path

Prompt AI → Read + understand

↓

Adapt and test incrementally

↓

It works. You understand why.

Skill level grows.

If your time is limited — go for the safer option. Learn when you have time.

Plan ahead. Don't save it for 4 am when the assessment is due at 7 am.

Research: anthropic.com/research/AI-assistance-coding-skills

Anthropic Research: AI Assistance & Coding Skills

Shen & Tamkin, 2026 — Randomised Controlled Trial — anthropic.com/research

52 junior engineers learned Trio, a Python library none had used before

Half used AI assistance, half coded manually — then both took a quiz

Without AI

67%

comprehension quiz score

(credit)

−17%

With AI

50%

comprehension quiz score

(barely pass)

No speed benefit

Only ~2 min faster

Not significant

Debugging hurt most

Gap: debugging,

code reading, concepts

HOW matters most

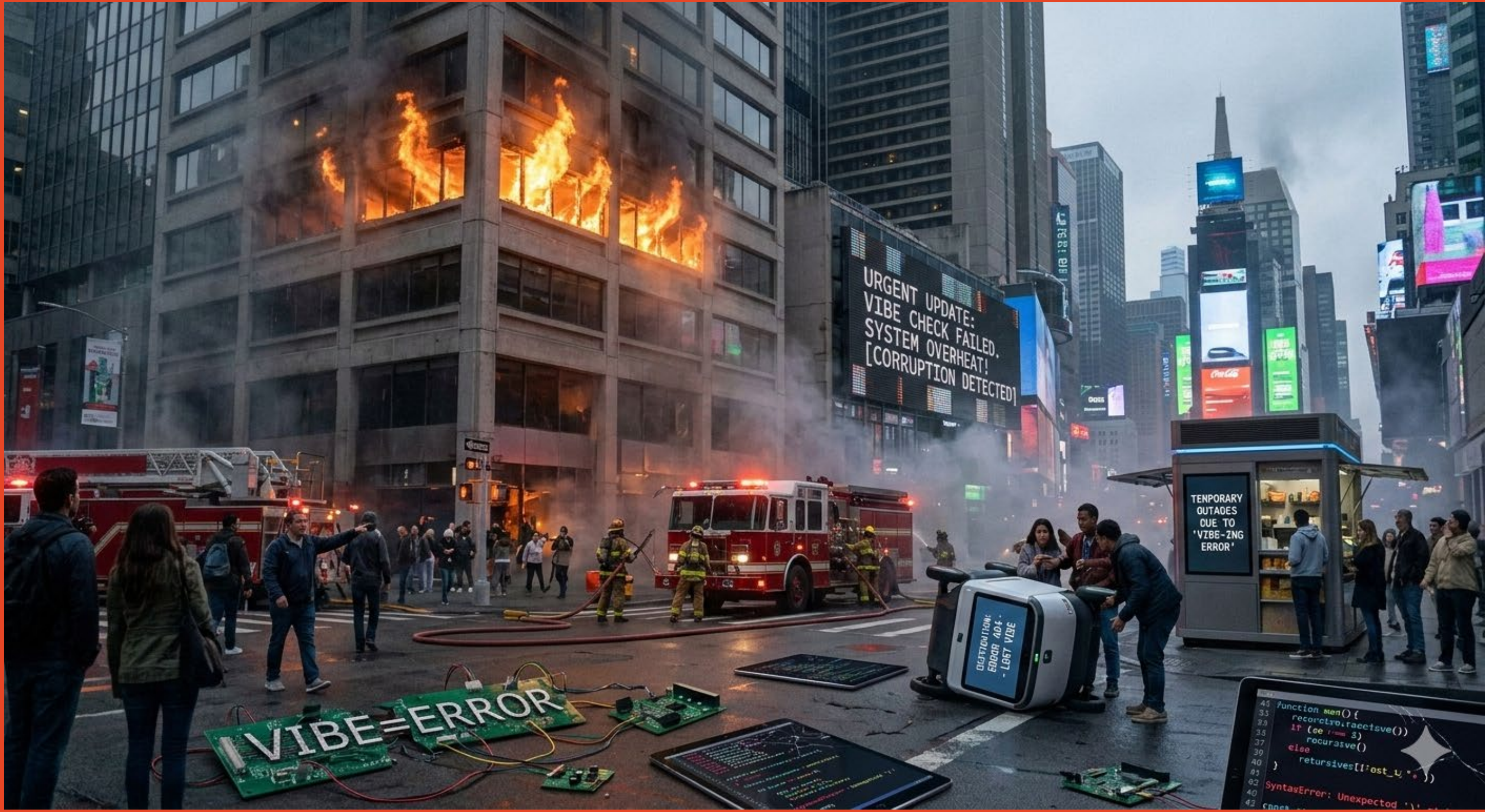
Ask to explain → 65%+

Ask to write → <40%

AI accelerates tasks where you already have skills.

When learning something new, use AI to understand — not to delegate.

Examples of Vibecoding Gone Wrong



BREAKING | BUSINESS

Samsung Bans ChatGPT Among Employees After Sensitive Code Leak

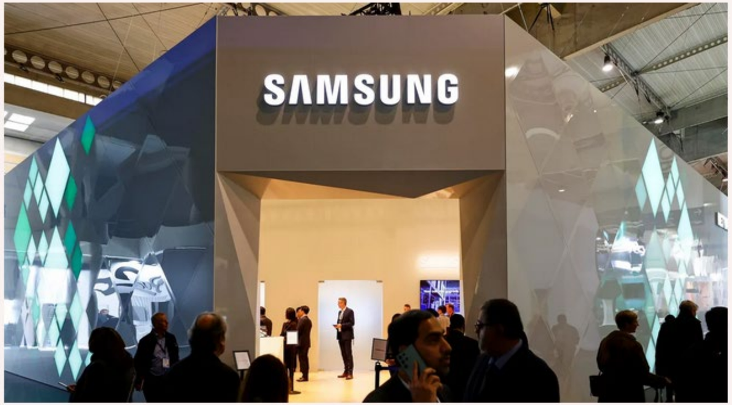
By Siladitya Ray, Forbes Staff. Siladitya Ray is a New Delhi-based Forbes news...

Follow Author

Published May 02, 2023, 07:17am EDT, Updated May 02, 2023, 07:31am EDT

TOPLINE

Samsung Electronics has banned the use of ChatGPT and other AI-powered chatbots by its employees, Bloomberg reported, becoming the latest company to crack down on the workplace use of AI services amid concerns about sensitive internal information being leaked on such platforms.



File Photo: Samsung Electronics has banned the use of ChatGPT and other similar tools at the workplace.
NURPHOTO VIA GETTY IMAGES

https://www.forbes.com/sites/siladityaray/2023/05/02/samsung-bans-chatgpt-and-other-chatbots-for-employees-after-sensitive-code-leak

Policy: Generative AI (e.g., ChatGPT) is banned

Ask Question

Asked 3 years, 2 months ago Modified 14 days ago Viewed 1.6m times

5402

Locked. Comments on this question have been disabled, but it is still accepting new answers and other interactions. Learn more.

> **Moderator Note:** This post has been locked to prevent comments because people have been using them for protracted debate and discussion (we've deleted over 300 comments on this post alone, not even including its answers).
>
> The comment lock is not meant to suppress discussion or prevent users from expressing their opinions. You are (as always) encouraged to vote on this post to express your agreement/disagreement. If you want to discuss this policy further, or suggest other related changes, please Ask a New Question and use the `ai-generated-content` tag.
>
> This question remains featured because that is still the best, most prominent, and only *permanent* way that we have to announce this policy site-wide.

All use of generative AI (e.g., ChatGPT[1] and other LLMs) is banned when posting content on Stack Overflow.

This includes "asking" the question to an AI generator then copy-pasting its output *as well as* using an AI generator to "reword" your answers.

Please see the Help Center article: What is this site's policy on content generated by generative artificial intelligence tools?

Overall, because the average rate of getting *correct* answers from ChatGPT and other generative AI technologies is too low, **the posting of content created by ChatGPT and other generative AI technologies is *substantially harmful* to the site and to users who are asking questions and looking for *correct* answers.**

The primary problem is that while the answers which ChatGPT and other generative AI technologies produce have a high rate of being incorrect, they typically *look like* the answers *might* be good and the answers are *very* easy to produce. There are also many people trying out

10 Nov 2025

Companies call back workers as AI fails to replace jobs

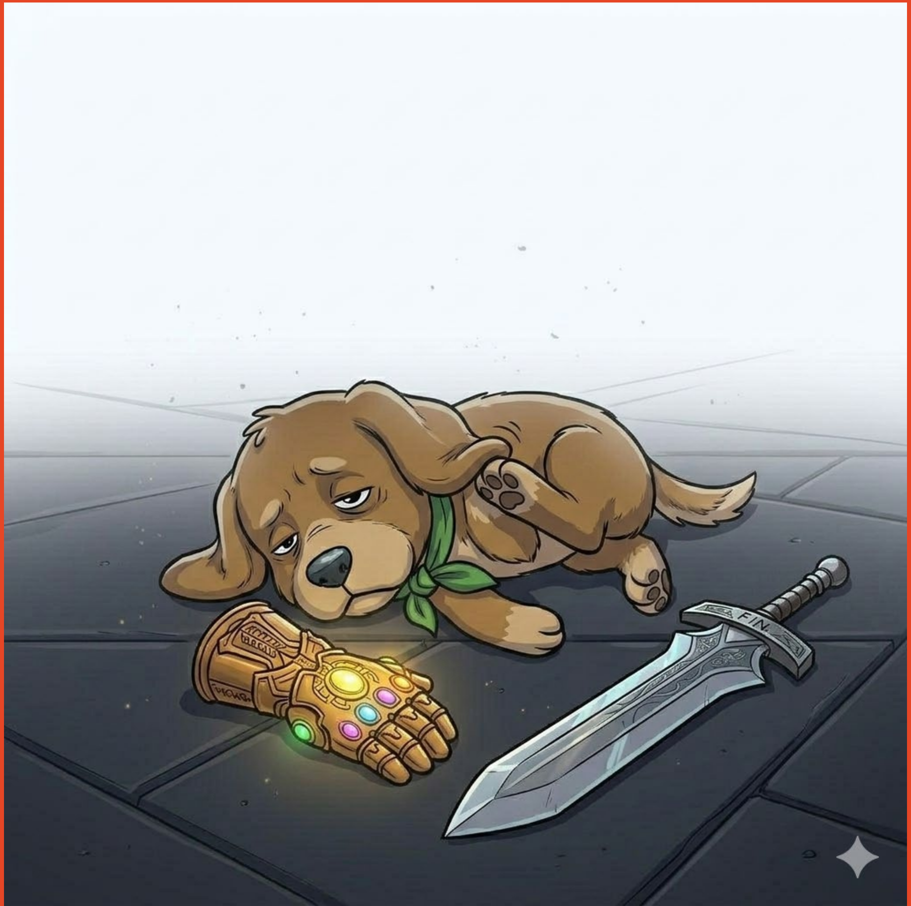
Around 95 percent of firms have yet to see measurable returns from AI, leading to rehiring trends.

Amazon’s cloud ‘hit by two outages caused by AI tools last year’

Reported issues at Amazon Web Services raise questions about firm’s use of artificial intelligence as it cuts staff

Amazon’s huge cloud computing arm reportedly experienced at least two outages caused by its own artificial intelligence tools, raising questions about the company’s embrace of AI as it lays off human employees.

Closing Thoughts



AI for coding is like Calculator to learn Maths

- AI is not bad. It is a tool, and you guys need to learn how to use it.
- Calculators are not bad, but if you want to learn maths, you need to learn first how to do the work without the calculator.
- It is up to you if you want to "cheat" and submit all the work generated by AI. I am here for those who care and want to learn the contents.


See you next week!
Go to your tutorials!
FIN